

A Complete Projective Atlas for Spheroidal Jeffery Directors in Planar Linear Flow

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Abstract

We classify a passive axisymmetric spheroid in every steady incompressible planar linear flow. The nonlinear head–tail director equation on $\mathbb{RP}^2$ is the projectivization of one traceless matrix exponential. A scalar discriminant then yields the complete elliptic, hyperbolic, nilpotent, and identity atlas. We prove the full source–saddle–sink stratification, the nilpotent fixed projective line, and the true head–tail versus oriented simple-shear periods. The result is source-local: clean periodic continua supply no arithmetic primitive-orbit owner or target determinant.

1 Frozen director and its linear lift

Let

$$L = \begin{pmatrix} a & b & 0 \\ c & -a & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad E = \frac{1}{2}(L + L^T), \quad W = \frac{1}{2}(L - L^T).$$

For a spheroid of finite aspect ratio $r > 0$, put

$$\lambda = \frac{r^2 - 1}{r^2 + 1} \in (-1, 1). \quad (1)$$

The head–tail director is $[p] \in \mathbb{RP}^2$. A unit representative obeys

$$\dot{p} = Wp + \lambda\{Ep - (p^T Ep)p\}. \quad (2)$$

For $r \neq 1$, $[p]$ is the intrinsic symmetry-axis director. An unmarked sphere ($r = 1$) has no intrinsic shape director; whenever $r = 1$ appears below, $[p]$ denotes a marked material director carried by the sphere. Jeffery [1] is cited only for the isolated-ellipsoid source equation, and Bretherton [2] only for rigid-particle and spheroidal-parameter lineage. No classification or proof below is outsourced.

Set $B = W + \lambda E$. Because $p^T W p = 0$, (2) is $\dot{p} = Bp - (p^T B p)p$.

Lemma 1 (Exact projective lift). *For every nonzero $q_0 \in \mathbb{R}^3$,*

$$[p(t)] = [e^{tB} q_0], \quad p(t) = \frac{e^{tB} q_0}{\|e^{tB} q_0\|}. \quad (3)$$

Conversely every solution of (2) has this form, so the flow is global on $\mathbb{RP}^2$.

Proof. If $q' = Bq$ and $p = q/\|q\|$, differentiation gives $p' = Bp - (p^T B p)p$. Conversely, multiplying a unit solution by $\exp \int_0^t p^T B p ds$ gives a solution of $q' = Bq$. Matrix exponentials are invertible, so normalization never vanishes. Passing to $[p]$ removes the representative sign exactly. $\square$

Write $k = (b - c)/2$ and $s = (b + c)/2$. The active block and its invariant are

$$B_2 = \begin{pmatrix} \lambda a & k + \lambda s \\ -k + \lambda s & -\lambda a \end{pmatrix}, \quad \delta = \lambda^2(a^2 + s^2) - k^2. \quad (4)$$

Direct multiplication gives $B_2^2 = \delta I$ and $\delta = -\det B_2$.

2 The complete sign atlas

Theorem 1 (All projective orbit types). *The sign and rank in (4) give exactly four cases.*

1. If $\delta < 0$ and $\omega = \sqrt{-\delta}$, the vertical director is the only fixed point. Every equatorial director has least period π/ω ; every director with nonzero horizontal and vertical parts has least period $2\pi/\omega$.
2. If $\delta > 0$ and $\rho = \sqrt{\delta}$, there are exactly three fixed eigen-directors $[v_+]$, $[e_0]$, $[v_-]$ with exponents $\rho, 0, -\rho$. Writing $q = q_+v_+ + q_0e_0 + q_-v_-$, its forward limit is $[v_+]$ if $q_+ \neq 0$, $[e_0]$ if $q_+ = 0, q_0 \neq 0$, and $[v_-]$ otherwise; backward time reverses $+$ and $-$. Thus they are sink, saddle, and source. Precisely,

$$W^s([e_0]) = \mathbb{P} \text{span}(e_0, v_-) \setminus \{[v_-]\}, \quad W^u([e_0]) = \mathbb{P} \text{span}(e_0, v_+) \setminus \{[v_+]\}.$$

Their closures are the two invariant $\mathbb{RP}^1$ projective lines.

3. If $\delta = 0$ but $B_2 \neq 0$, the fixed set is the entire projective line $\mathbb{P}(\ker B)$. Every other trajectory converges as $t \rightarrow \pm\infty$ to the unique horizontal line $\mathbb{P}(\text{im } B_2) = \mathbb{P}(\ker B_2)$, at algebraic rate $|t|^{-1}$.
4. If $B_2 = 0$, the flow is the identity on all of $\mathbb{RP}^2$.

Proof. Cayley–Hamilton gives, respectively,

$$e^{tB_2} = \cos(\omega t)I + \frac{\sin(\omega t)}{\omega}B_2, \quad e^{tB_2} = \cosh(\rho t)I + \frac{\sinh(\rho t)}{\rho}B_2. \quad (5)$$

In the first case there is no real horizontal eigenline. At π/ω the horizontal block is $-I$, which returns every equatorial head–tail line but not a mixed line; at $2\pi/\omega$ the full matrix is I . No earlier return is possible because a non-scalar elliptic block has no real eigenline.

In the second case, exponentiating the three eigen-coordinates gives the stated limits. Within $\mathbb{P} \text{span}(e_0, v_-)$, every point except $[v_-]$ tends forward to $[e_0]$; within $\mathbb{P} \text{span}(e_0, v_+)$, every point except $[v_+]$ tends backward to $[e_0]$. These are projective lines, and a point outside them has both corresponding leading coordinates, proving the stable/unstable completeness. Coordinate ratios give gaps ρ or 2ρ . At $\delta = 0$, $e^{tB} = I + tB$. A nonzero nilpotent 2 by 2 block has rank one and image equal to kernel. Hence $\mathbb{P}(\ker B)$ is fixed, whereas $[q + tBq] \rightarrow [Bq]$ off that line. Finally $B_2 = 0$ and the zero vertical block give $B = 0$. $\square$

3 Simple shear and singular shape faces

For $L_2 = \begin{pmatrix} 0 & \dot{\gamma} \\ 0 & 0 \end{pmatrix}$ with $\dot{\gamma} \neq 0$, substitution of (1) in (4) yields

$$\delta = -\frac{\dot{\gamma}^2 r^2}{(r^2 + 1)^2}, \quad \omega = \frac{|\dot{\gamma}|r}{r^2 + 1}. \quad (6)$$

Thus the physical head–tail equatorial period and the oriented-vector period are

$$T_{\mathbb{RP}^1} = \frac{\pi(r + r^{-1})}{|\dot{\gamma}|}, \quad T_{\text{oriented}} = \frac{2\pi(r + r^{-1})}{|\dot{\gamma}|}. \quad (7)$$

The second value is the least period of every nonvertical oriented vector and also of a mixed $\mathbb{RP}^2$ director; the vertical oriented vector is fixed. This factor two is a state-space fact, not a convention change.

At $r = 1$, $\lambda = 0$ and strain drops out: under the marked-material-director convention above, the marked direction follows only vorticity. The unmarked sphere itself has no observable orientation state. As $r \rightarrow \infty$ or $r \rightarrow 0$, $\lambda \rightarrow \pm 1$, the simple-shear generator is nilpotent and (7) diverges. These rod/disk limits are not finite-aspect periodic points. At $\dot{\gamma} = 0$, every director is fixed.

4 Executable receipt and claim boundary

The deterministic certificate contains 625 exact parameter rows, 320 90-digit orbit reconstructions, ten shear-period rows, five strobe rows, and six boundary rows. The producer-independent checker passes 8,328 assertions; the

symbolic reconstruction passes 39 identities; replay is byte-exact; and 25/25 repaired-hash or stale-hash hostile mutations are rejected. These rows are regression oracles, not a finite proof of the arbitrary-parameter theorem.

The honest Route-A tuple is

$$(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL), \quad \text{ROUTE_A_REJECTED}. \quad (9)$$

There is no rational-prime carrier or logarithmic clock. Elliptic periodic points occur in clean continua with continuously variable periods; the other chambers align or are stationary. Consequently no isolated primitive-orbit product, target determinant, target zero match, Hilbert–Pólya operator, or Route-B input follows. The locked scope is NO_BAD_EULER_OR_ROOT_NUMBER.

References

- [1] G. B. Jeffery, “The motion of ellipsoidal particles immersed in a viscous fluid,” *Proc. R. Soc. Lond. A* **102** (715), 161–179 (1922), doi:10.1098/rspa.1922.0078.
- [2] F. P. Bretherton, “The motion of rigid particles in a shear flow at low Reynolds number,” *J. Fluid Mech.* **14** (2), 284–304 (1962), doi:10.1017/S002211206200124X.